



Blockchain-enabled federated learning for prevention of power terminals threats in IoT environment using edge zero-trust model

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Abstract

The continuous deepening of information technology in the power industry has dramatically increased the exposure of the power Internet of things. Attackers use the compromised terminal as a springboard to infiltrate the network and can steal sensitive data in the power industry system or carry out damage. Facing the bottleneck of zero-trust centralized deployment of massive power terminal access, an edge zero-trust model is proposed. A cross-context recommendation strategy based on self-attention-based federated learning (SAFL) is proposed in this work. In contrast to current algorithms, SAFL fully mixes the target and auxiliary background information. FL is a technological solution that permits machine learning on-device without transferring the user's private data to a centralized cloud. Therefore, federated learning can help to accomplish personalization. Around the dumb power terminals, the zero-trust engine is deployed in a distributed multi-point, and the trust factors are collected in real-time and stored on the chain. By maintaining a consortium blockchain, the trust factor blockchain (TF_chain), the storage-type edge server synchronizes and shares the trust factor generated by the power terminal during the movement which is convenient for tracing the source and preventing the information from being tampered with abnormal and sensitive elements. Furthermore, lightweight encryption is adopted to ensure authentication information is transmitted from edge to cloud security. The simulation results show that the proposed model can decentralize the zero-trust processing load of centralized deployment and effectively combat the threat of compromised terminals under the condition of marginalized deployment.

Keywords Blockchain · IoT · Edge computing · Zero-trust model · Power terminal

1 Introduction

Internet of things' potential is an industrial-grade Internet of things applied to the power industry, which can quickly improve the perception, interaction, and adjustment capabilities of the power grid and ensure the safe, stable, continuous, and efficient operation of the "generating, transmission, transformation, distribution, and utilization" links of the power system. [1]. Power transmission, distribution, and generation are the three phases of a power grid. The details of each of these phases are provided below. A network of synchronized power producers and consumers, connected by transmission and distribution lines, and managed by one or more control centers makes up an electric grid. The force Internet of things is an industrial-grade Internet of things used in the power sector, which can quickly enhance the perception, interaction, and adjustment capabilities of the power grid and guarantee the "generating, transmission, transformation, distribution, and utilization" links of the power system operate safely, steadily, continuously, and efficiently. As a critical infrastructure related to the national economy and people's livelihood the power Internet of things has always been a critical target of network attacks. With the wide application of emerging technologies such as "Big Cloud, IoT, Smart Chain Edge" in the power industry, the types and number of terminals connected to the power Internet of things have increased significantly [2]. A network receives real-time data from sensors and devices at the Internet of things (IoT) edge. Since data are processed closer to its point of origin using IoT edge computing, latency problems with the cloud are resolved. Predictive maintenance of a range of assets using condition monitoring is the most popular use in the power sector. The IoT-based strategy shifts away from conventional reactive and routine maintenance tactics and toward proactive ones. Applications are concentrated on the most valuable assets in the grids for transmission and distribution as well as in generating facilities. Accessing to many terminals further increases the exposure of the power Internet of things, which poses a severe challenge to the network security defense system characterized by boundary isolation [3]. Studies have shown that most smart terminals have security risks and vulnerabilities, and the firmware of related terminals also has backdoors implanted by manufacturers [4]. The security of the intelligent terminal determines the security and stability of the power Internet of things to a great extent [5]. Once attacked, it may lose the average control ability of the power terminal [6]. Attackers can locate weak power connectors for identity theft or incursion control. They can employ internal threats using the compromised terminal after getting past the border network security protection system. It is simple to create bogus data and send fraudulent control orders once the attacker has gained access to the power network through a springboard. Once the attacker invades the power network through a springboard, it is straightforward to forge false data and send malicious control commands [7]. A machine learning technique called federated learning enables numerous devices to train a single shared model without sharing raw data. Numerous devices can train a single shared model using federated learning, a machine learning technique, without exchanging raw data. Federated learning, which is comparable to a

group presentation or report, involves the remote sharing of data by a number of individuals in order to train a single deep learning model jointly and frequently. A recently developed technique called federated learning (FL) has piqued the curiosity of researchers who are eager to learn more about its potential and utility. Federated learning, which is like a group presentation or report, entails the remote sharing of data by several people in order to train a single deep learning model collectively and repeatedly. Researchers are very interested in investigating the potential and usefulness of federated learning (FL), a recently established technology. The algorithms being used may not always gain experience when using traditional machine learning techniques. FL makes it possible. FL has been applied to numerous industries, including mobile apps, the Internet of things, transportation, and defense. Accessing several terminals increases the power Internet of things' exposure, posing a serious challenge to the network security defense system's border isolation principle. IoT devices, such as sensors, are used for monitoring, data collection, and automation in sectors including manufacturing, logistics, and agriculture. This enables increased operational efficiency, predictive maintenance, and process optimization. Multiple data owners (referred to as clients) collaborate to train a single machine learning model using federated learning (FL), a novel learning paradigm, without sharing their own training data.

The "2020 Cost of Insider Threats: A Global Report" [8] released by the Ponemon Institute shows that the cost of data breaches caused by insider threats has increased by 31% in two years, reaching more than \$10 million. Existing insider threat detection methods focus on malicious insider detection [9]. However, the internal threat caused by the lost terminal may be more harmful, and the main reason lies in winning by quantity and sudden behavior change. The internal threat brought on by the lost terminal, however, might be more dangerous, and the primary causes are quantity winning and abrupt behavior changes. Due to the large number and variety of power terminals, the better the attack opportunity to accomplish the goal of winning by quantity of the lost terminal has been engaged in until the intrusion does not control it is the more the attackers control the lost terminals by casting a wide net. Due to the large number and types of power terminals, the more the attackers control the lost terminals by widely casting a net, the more excellent the attack opportunity to achieve the goal of winning by quantity of the lost terminal has been engaged in normal behavior until the intrusion does not control it. Attackers can suddenly do bad things under the guise of "legitimate identity," and this behavioral mutation can fully use the time lag of attack detection to win by surprise.

Following the principle of "never trust, always authenticate," all devices, users and traffic in the network should be authenticated and authorized. Zero trust has become an effective means to combat internal threats [10]. However, the existing research schemes focus on the centralized deployment of zero-trust engines. Smart terminals include security flaws and hazards, and manufacturers have inserted backdoors into the firmware of linked terminals. Paper receipts are no longer required, cloud storage is possible, many payment methods are accepted, and a higher level of security is offered by smart terminals. Therefore, it is challenging to meet the authentication requirements in the scenario of massive terminals accessing the power IoT. In addition, if the deployment of the zero-trust engine is too far from the

power terminal, it is easy to delay the authentication time, which is not conducive to applying zero-trust in the delay-sensitive power IoT scenario. Federated learning is a promising strategy for edge intelligence with privacy protection in distributed environments. The acquisition of images, pre-processing, feature extraction, and classification are the typical phases included in the framework of CML techniques for image classification. Most of the time, machine learning algorithms excel with modest datasets. In contrast to conventional machine learning algorithms, deep learning takes a lot of data to understand and execute well. In contrast to conventional machine learning, which collects all training data at a single curator, federated learning distributes the training effort across users, which significantly minimizes data transmission costs. Users in a framework for federated learning store their data on their own devices but send the parameters to the server for aggregate. This offers a complementary method for users to jointly understand a global model of data privacy. Federated learning uses blockchain to create a guaranteed collaboration system among untrusted players for effective sharing, which enables learning from distributed data while maintaining anonymity.

According to a detailed analysis of the authentication requirements in the scenario of widespread terminal access to the power Internet of things, a zero-trust engine should be deployed around the dense power terminals to combat the potential threat of compromised terminals.

This may help to create an edge zero trust model. This paper proposes an edge zero-trust model against the threat of lost terminals for the power IoT scenario of massive terminal access and introduces three key innovations:

1. The internal monitoring and authentication functions of the network endowed by zero trust are sunk to the edge of the power Internet of things, close to the power terminal, and the zero-trust engine is deployed in a distributed multi-point to prevent single-point failure and denial of service attacks. Furthermore, the cooperation operation mechanism between zero trust core components is clarified, and virtualization configures the zero trust core components by reducing-edge servers' consumption.
2. Edge separation trust evaluation involves the trust factor's collection, storage, and computation. Make full use of the zero-trust client installed on the power terminal, collect trust factors in real-time, and submit them to the nearest edge server. The alliance blockchain is introduced, the trust factors chain (TF_chain), which is jointly maintained by the storage edge server to realize the sharing of trust factors generated by the power terminal during the movement, which helps to trace the source and prevent the information from being tampered with the zero-trust engine is only responsible for authentication and scheduling the data. The TF_chain and the trust value of the power terminal can be quickly calculated, which avoids the data migration that is prone to delay during the replacement of the zero-trust engine. Furthermore, abnormal and sensitive factors are extracted from the trust factors to form an edge dynamic trust evaluation scheme, which can quickly respond to the behavioral mutation of the trapped terminal and block it in time.

3. A signcryption scheme suitable for edge authentication information transmission is designed, and the certificate strategy based on an elliptic curve avoids the critical escrow problem. This scheme has good lightness in computing efficiency, can resist the tampering of edge authentication information, and ensure safe transmission to the cloud. After that, the power data center in the cloud only needs to verify and issue authorization credentials and provide related services according to the resource access request type, which relieves the pressure of security authentication.

This paper appears to be more computationally efficient. Because of this, the encryption method presented in this research can guarantee the secure transmission of edge authentication data while also having greater computing efficiency and lightweight. The simulation outcomes demonstrate the hypothesized edge of zero-confidence. The arbitrary model has a higher lost terminal detection rate and effectively reduces the harm posed by lost terminals.

2 Related work

With the gradual weakening of the network security boundary, zero trust has been paid more and more attention by the industry. Since the Google Aurora incident, Google has started a research program in the field of zero trust. Beyond Corp [11] architecture for employees has become the first implementation of zero trust, which clearly defines the security between users' devices and applications. A trusted chain is established through the continuous authentication mode, allowing legitimate users to access protected services. A technology advancement called FL enables machine learning on-device without sending the user's personal information to a centralized cloud. Federated learning can therefore aid in achieving personalization. The zero-trust engine is installed in a dispersed multi-point configuration around the dumb power terminals, and the trust factors are gathered in real-time and saved on the chain. Federated learning is a decentralized method for training machine learning models (also known as collaborative learning). Data transfer from client devices to global servers is not necessary. Tencent summarizes the core idea of zero trust in the "Zero Trust Solution White Paper" [12], pointing out that access subjects are not distinguished from inside and outside the network, and unverified traffic is untrusted by default. Furthermore, in the "Guiding Opinions on Promoting the Development of the Network Security Industry (Draft for Comment)," the Ministry of Industry and Information Technology listed zero-trust security for the first time as a "key technology for network security that needs to be broken through" [13]. Many users can cooperatively train an ML model without releasing their personal information thanks to federated learning (FL), a revolutionary distributed machine learning (ML) approach. In a typical FL system, a central server manages the global model and regulates the users. Federated learning is used to keep knowledge from being altered with or exposed to unusual or sensitive aspects. Encryption, access restrictions, and data masking are often used methods to ensure confidentiality. Integrity is crucial to ensuring that data has not been altered or interfered with without authorization. This

covers guarding data against illegal addition, modification, or destruction. Every single client sends the local model updates to the central server for rapid aggregation while keeping the raw data in their local storage. Since no sensitive information is transferred during the training process, FL provides a strong privacy guarantee to the collaborating clients and has several applications in edge computing, finance, and healthcare.

The “Zero Trust Security Architecture” draft standard released by the National Institute of Standards and Technology (NIST) [14] pointed out that zero-trust security architecture is an end-to-end network/data protection method, including identity, Credentials, access management, operations, terminals, host environments, and interconnected infrastructure. As the most authoritative zero-trust security architecture at present, the architecture proposed by NIST mainly includes subjects, interviewed resources, auxiliary systems and zero-trust engines, as shown in Fig. 1. The subject is the user, device or terminal that can access resources within the network. Usually, a zero-trust client is installed on the main body for risk perception, and the device’s security status is monitored in real-time. In time, the zero-trust client will detect suspicious behaviors such as unauthorized operation and malicious data injection, and the zero-trust gateway will be informed to intercept. However, suppose the zero-trust client is not installed, or the manipulation of the zero-trust client occurs. In that case, the subject cannot conduct communication activities and request access to resources in the network content.

Because of the risk-awareness mechanism of zero-trust clients that can block suspicious behaviors in abnormal traffic in time, the zero-trust core components, such as policy engine (PE), policy administrator (PA), policy enforcement point (PEP), etc., the trust engine focuses on protecting the visited resources inside the network and preventing the threat of sensitive information from being stolen. Among them, the PE analyzes the credibility of the subject to form the authentication result; the PA decides to allow or reject the connection according to the authentication result; the PEP is responsible for continuous monitoring and ends the connection between the subject and the visited resource and is usually called a zero-trust gateway. Zero trust can effectively improve the defense capability of the data center against lateral movement of malicious nodes in the system and network penetration attacks

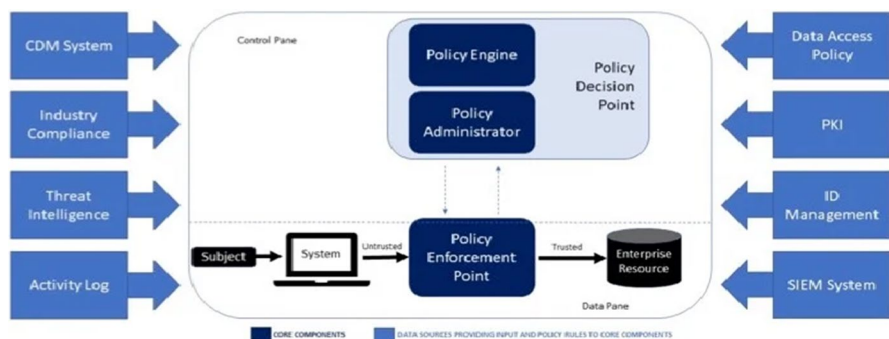


Fig. 1 Zero-trust security architecture proposed by NIST

[15]. Therefore, zero trust has quickly become a research hotspot in academia, and researchers mainly rely on the zero-trust security architecture proposed by NIST to carry out related research.

Reference [16] proposed an access control strategy based on the zero-trust network for the problem of MAC spoofing in the software-defined network framework of cloud architecture. Reference [17] offers a zero-trust-based IoT security architecture, which adopts the strategy of comprehensive authentication and real-time monitoring to realize the security of the power IoT. Reference [18] proposed a conceptual zero-trust approach in the cloud environment to prevent the risk of data leakage and ensure the integrity and confidentiality of data. Reference [19] designed a security protection framework based on zero trust from two aspects of identity recognition and access control, aiming at the security risks faced by the current electric mobile interconnection. Maintaining desired performance and stabilizing the power system in the wake of various disruptions, such as short circuits and loss of generation and/or load, are referred to as power system control. The ability of a power system to resume normal or steady performance following exposure to disturbances is referred to as stability.

Reference [20] proposes a defense system based on the zero-trust model for medical and health data vulnerabilities. Only authenticated users and devices can interact with the network to ensure the security of data transmission. Researchers in the FL community have also offered various defense strategies versus adversarial examples. However, these defenses are useless against undirected Byzantine attacks because they were originally developed for attacks that were deliberately targeted. Parallel learning, distributed learning, and deep learning are typically contrasted with FL. Although FL is a relatively new field, it has been extensively examined in several publications that are relevant to FL [21].

In the zero-trust security architecture, the authentication of entities is mostly based on trust evaluation. At present, there are already few trust evaluation mechanisms. Based on the research status at home and abroad, the deployment of zero-trust engines tends to be centralized. The comprehensive research status at home and overseas can be effectively monitored in the scenario with a limited number of terminals. Therefore, the deployment of zero-trust engines tends to be centralized. The system's update that can be effectively monitored lags behind the change in the subject's behavior. It is challenging to deal with the terminal threat of behavior mutation. Table 1 shows the advantages of our model over the latest related research work.

3 System architecture

As an essential technology of the power Internet of things, edge computing is essentially a distributed computing architecture. By sinking cloud computing and storage capabilities to the network edge, localized, close-range, and distributed applications, services, and content deployment can be achieved. In addition, localized distributed security protection has become essential to ensure the stable operation of the power Internet of things.

Table 1 Comparison with related works

Related work	Introducing Zero Trust	Monitoring method	Trust assessment	Sudden behavior block	Authentication information transfer	Trust factor collection
Literature [17]	Support	Centralized	Not support	Not supported	Encryption	Not support
Literature [18]	Support	Centralized	Dynamic	Not support	Not support	Not support
Literature [19]	Support	Centralized	Static	Not support	Encryption	Centralized collection
Literature [20]	Support	Blockchain	Not support	Not support	Not support	Not support
Model of this paper	Support	Marginalized	Dynamic	Block in time	Signerption	Real-time collection at the edge

Because of this, the internal monitoring and authentication functions of the network endowed by zero trust can be sunk to each edge area divided according to the “generating, transmission, transformation, distribution, and utilization” links of the power system, and the zero-trust engine can be implemented around the power terminal: distributed deployment and associated trust assessment. Several centralized data poisoning strategies can be used right away to cause FL poisoning. In federated learning, the attacker can directly manipulate model updates to increase attack performance, in contrast to data poisoning in centralized learning. Multi-party on-device learning is made possible by federated ML, which splits the effort [22]. Due to federated learning’s recent development, it is still unclear if it can be utilized to improve anomaly detection in scenarios when resources are few and what impact this will have on the achieved performance and accuracy trade-off. The edge zero-trust model in the power IoT scenario can be implemented on the “end,” “edge,” and “cloud” system architecture. As shown in Fig. 2, the architecture includes a terminal layer, edge layer and cloud layer.

1. Terminal layer. A zero-trust client is installed on each networked power terminal to monitor its network activities and resource access requests in real-time. The risk perception mechanism of the zero-trust client can block suspicious behaviors in abnormal traffic in time. However, for the resource access request of the power terminal, it is generally difficult for the zero-trust client to determine the doubtful situation, and it needs to be submitted to the zero-trust engine for authentication and judgment.
2. Edge layer. According to the business scope, the power Internet of things can be divided into f edge regions, defined as the set $\Gamma = \{EA_1, EA_2, \dots, EA_k, \dots, EA_f\}$. In each area, uniformly distribute edge services the set of devices $\Lambda = \{\theta,$

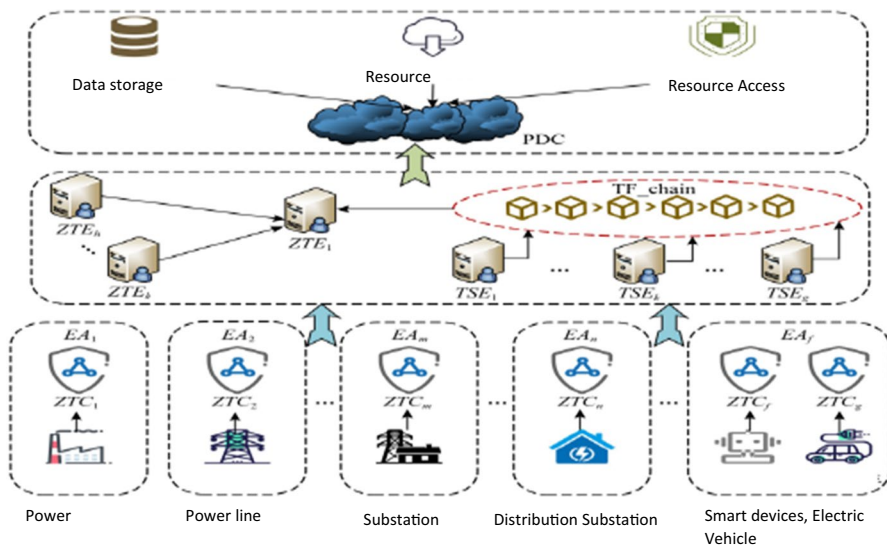


Fig. 2 Network architecture of edge zero-G-trust model

Φ }. Among them, $\Theta = \{\text{TES}_1, \text{TES}_2, \dots, \text{TES}_k, \dots, \text{TES}_g\}$ is a set of trust storage engines composed of g edge servers, which jointly maintain a TF_chain to achieve traceable and tamper-proof terminal trust factor storage; $\Phi = \{\text{ZTE}_1, \text{ZTE}_2, \dots, \text{ZTE}_k, \dots, \text{ZTE}_h\}$ is a set of zero-trust engines composed of h edge servers, including a primary engine and $h-1$ secondary engines. When the main engine is down, it automatically upgrades a secondary machine to the main engine. An edge zero-trust paradigm is suggested in order to address the bottleneck of zero-trust centralized deployment of enormous power terminal access. This article proposes a self-attention-based federated learning (SAFL) cross-context recommendation technique. Unlike other algorithms, SAFL fully combines the target and auxiliary background data. FL is a technology advancement that enables on-device machine learning without sending user data to a centralized cloud. Consequently, federated learning can aid in achieving personalization. Each ZTE edge server loaded with a zero-trust engine is virtualized according to its hardware and software resources. It is virtualized into three virtual machines that perform zero-trust core components such as PE, PA, and PEP. As a privacy-preserving model for distributed edge intelligence, federated learning develops. In order to provide ultra-reliable low-latency communication in automotive networks, a joint transmit power and resource allocation technique was developed. The federated learning technique has been enhanced with control algorithms in several research. The asynchronous federated learning architecture being suggested is empowered by blockchain.

3. Clouds. Any process such as power transmission, distribution, and sales require a robust data center to serve it. After sinking the monitoring and authentication functions of zero trust to the network edge, the power data center (PDC) only needs to verify and issue authorization credentials and provide complementary services according to the type of resource access request. Under the premise, the certification efficiency is effectively improved.

4 Edge zero-trust model design

In this section, an edge zero-trust model is constructed to adapt to the monitoring and authentication requirements of massive terminals in the power IoT scenario and to achieve fast and effective detection of lost terminals.

4.1 Edge zero-trust engine deployment

Distributed deployment of the zero-trust engine in each edge area is the first step in building an edge zero-trust model. With virtualization, zero-trust core components such as PE, PA, and PEP can be configured on three virtual machines on a single edge server without spending on three edge servers. Servers are configured separately. The cooperation between the zero-trust core components constitutes the operation mechanism of the zero-trust model, which mainly includes six steps:

- 1 The resource access request sent by the power terminal will be generated by the zero-trust client and reported to the PEP;
- 2 PEP first intercepts resource access requests and then forwards the authentication requirements to PE;
- 3 PE calls the trust factor stored on TF_chain, calculates the terminal trust value, authenticates resource access requests, and distributes the authentication results to PA and PEP;
- 4 For the subject that passes the authentication, the PA generates the authorization credential, and sends it to the PEP;
- 5 The PEP submits the authentication result and authorization credential to the PDC of the cloud layer;
- 6 After verifying that the authorization result is valid, the PDC notifies the PEP to release the interception of the resource access request.

Even if the zero-trust engine is deployed at the edge, if the access requests sent by the power terminal in a particular edge area are frequent, the zero-trust engine may be down. To ensure that authentication data is sent from edge to cloud security, lightweight encryption is used. The simulation results demonstrate that the suggested architecture can successfully counter the threat of compromised terminals under the circumstance of marginalized deployment and decentralize the zero-trust processing load of centralized deployment. In particular, the attacker can control some compromised terminals to generate a large amount of regular traffic without any malicious code injected without knowing the location of the zero-trust engine and automatically flow to the zero-trust machine through the zero-trust client to form a denial-of-service attack.

In this regard, the distributed multi-point deployment of the zero-trust engine should be implemented in each edge area to form a set of zero-trust machines $\Phi = \{ZTE_1, ZTE_2, \dots, ZTE_k, \dots, ZTE_h\}$. The rules of edge distributed multi-point deployment are as follows:

Rule 1. The h zero-trust engines in the Φ set are scattered in different locations of each edge region to avoid location detection by attackers. The main engine is working and responsible for monitoring the power terminals in the entire edge area. The remaining $h-1$ secondary machines are on standby and establish the rotating secondary machine set Ξ , waiting to become the main engine.

Rule 2. Assuming that ZTE_k is the main engine, ck represents its processing capacity. Set the alarm threshold of processing capability to δ . Once $ck \geq \delta$, ZTE_k broadcasts an offline notification to $h-1$ secondary engines in the Ξ set.

Rule 3. The main engine is in a high-intensity working state for a long time, which may also cause pressure overload to the edge server. Therefore, the rotation time of the given ZTE_k is t_k , and the maximum rotation time is set as τ . When $t_k \geq \tau$, ZTE_k broadcasts an offline notification to $h-1$ secondary engines in the Ξ set.

Rule 4. To avoid the failure of the main engine to send offline notifications due to sudden downtime, the first-ranked secondary engine in the Ξ set (assumed to be ZTE_p) needs to monitor the activity status of the main engine all the time.

Rule 5. If ZTE_p detects the sudden downtime of ZTE_k , it broadcasts the offline notification of ZTE_k and the online message of becoming the main engine to the secondary engines in the Ξ set.

Rule 6. After the downed main engine is recovered, it is loaded into the Ξ set and becomes the last secondary engine.

4.2 Edge dynamic trust evaluation

Dynamic trust assessment is the core link of the zero-trust model to achieve rapid authentication of power terminals. With the edge deployment of the zero-trust engine, the edge dynamic trust evaluation scheme is timely proposed to help reduce the cloud computing load. Combining local customer models created using federated learning (FL), a machine learning (ML) technique, and related architecture, a server learns a machine learning (ML) model. Federated learning as the method of carrying out a learning task via a loose network of participating devices that are controlled by a central server. Training a deep learning model cooperatively with hundreds of people is now possible thanks to a novel concept called federated learning. The primary goal of federated learning is to build a cooperative machine learning model using local datasets while ensuring privacy. Many forthcoming scenarios, such as edge computing and crowdsourcing systems, are attractive candidates for this approach. As shown in Fig. 3, the implementation of the edge dynamic trust evaluation scheme involves the collection of trust factors and the storage on the chain and the calculation of the terminal trust value for the PE calling service.

4.2.1 Trust factor collection and on-chain storage

In monitoring the network activities of the power terminal, the zero-trust client needs to collect the trust factors in real-time according to the changes in terminal behavior and submit them to the nearest TSE edge server for storage on the chain.

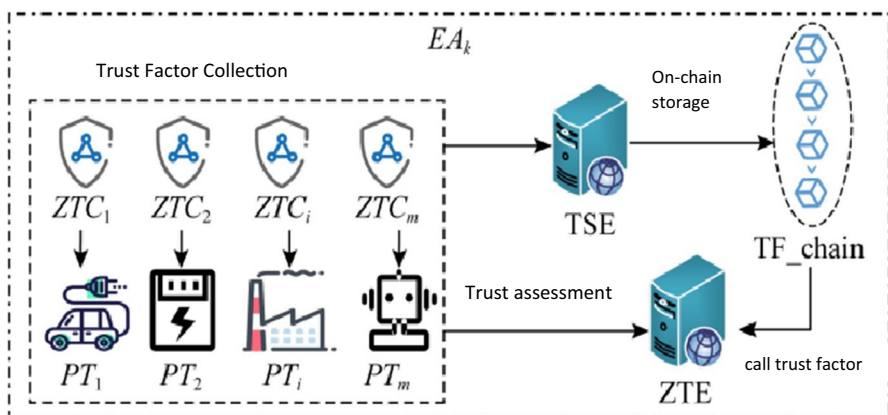


Fig. 3 Edge dynamic trust evaluation scheme

Define the power terminal set $\Delta = \{PT_1, PT_2, \dots, PT_i, \dots, PT_m\}$ in the edge area EA_k . Taking PT_i as an example, ZTC_i represents the zero-trust client installed on it. The trust factor set of PT_i contains three types: BTF, ATF, STF, and typical trust factors. The factors are listed in Table 2. BTF represents the essential trust factor, which is used to calculate the critical trust value of PT_i ; ATF represents the abnormal trust factor, which is used to attenuate the critical trust value of PT_i ; STF represents the sensitive trust factor, which is used for timely blocking PT_i may be out of control of the situation.

Whenever a specific trust factor of PT_i changes, ZTC_i sends updated information to the nearest TSE edge server in real-time. Because of the characteristics of the pre-selected consensus miners of the alliance blockchain, the TSE edge server nearby ZTC_i serves as the consensus miner head of the current TF_chain blockchain. Every time a trust factor changes, a new block is generated.

To improve the block structure of Bitcoin, based on inheriting the original parameters {previous block hash, block ID, timestamp, Merkle root}, add $\{PT_i, sn_i^p, r_i\}$ to the block header, which packs the updated trust factor into the block body, as shown in Fig. 4.

In the block header, sn_i^p represents the sequence number of the newly generated block B_IDp about PT_i on the TF_chain blockchain, and r_i is the amount of redundancy. The serial number starts from 1, and Y represents the set of all serial numbers on the TF_chain blockchain. Each block of PT_i is stored in the block body trust factor.

Because the edge area is divided according to the “transmission, transmission, transformation, distribution, and use” links, the movement of power terminals generally does not exceed this area. However, in real-time, ZTC_i sends PT_i trust factor updates information to the nearest TSE edge server. Every time a trust factor update is sent, a new block will be generated, which cause a large number of redundant blocks, and increase the storage load of each TSE edge server. In this regard, combined with the timing characteristics of the blockchain and the block storage structure of the trust factor, the Algorithm 1 eliminates redundant blocks. Algorithm 1 can be embedded in the smart contract of the TF_chain blockchain to run

Table 2 Trust factors

Type	Identifier	Properties
BTF	btfi1	Authentication succeeded
	btfi2	Authentication failed
ATF	atfi1	Password exception
	atfi2	Request exception
	atfi3	Information detection
	atfi4	Packet exception
STF	stfi1	Identity fraud
	stfi2	IP address is misplaced
	stfi3	Unauthorized request
	stfi4	Information tampering

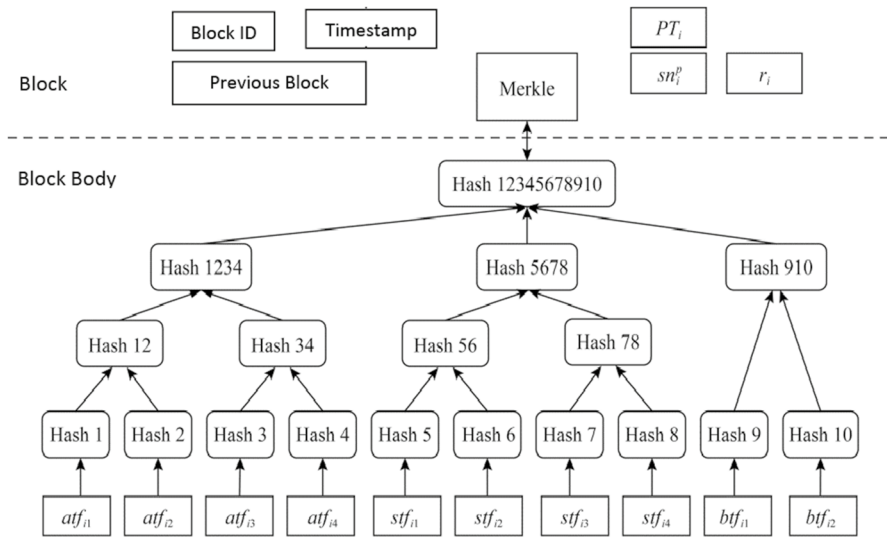


Fig. 4 Block storage structure for trust factors

automatically regularly to eliminate redundant blocks forward, which is a TSE edge. Edge server to reduce storage pressure.

The biggest advantage of trust factors stored on the TF_chain blockchain is tamper resistance. Blockchain is a chain storage structure, the next block saves the hash value of the previous block. If an attacker tampers with one of the blocks, all the blocks following the block must be modified. Assuming that there are l_p blocks behind the block B_ID_p after tampering with the block, the number of blocks Z_p that needs to be tampered with is computed as follows:

$$Z_p = g + (g)^2 + \dots + (g)^{l_p} = \frac{(g)^{l_p+1} - g}{g - 1} \quad (1)$$

For example, when the number of TSE edge servers $g=5$, $l_p=6$, $Z_p=19530$, so the TF_chain blockchain can hardly be tampered with.

4.2.2 Calculation of terminal trust value

Beta distribution can fit well with trust distribution, and its mathematical expectation can be used to calculate the trust value of nodes. It is one of the most classic and widely used trust evaluation models. Beta distribution has the characteristics of simple calculation, good flexibility, strong statistical ability and strong applicability. Node selection selects the federated learning nodes with the highest number of resources in the allotted communication time in order to increase running efficiency and training accuracy. The chosen nodes serve as the permissioned blockchain's verifiers as well. The server starts the federated learning operations by selecting an overall machine learning model and setting initialization settings. Therefore, take

Input: $\Delta, Y, \tau_i, TF_chain$;

Output: TF_chain .

- 1) for $PT_i \in \Delta$ then
 - 2) if $\tau_i \geq 1$ then
 - 3) if $sn_i^p = \min(Y)$ then
 - 4) Eliminate the current redundant blocks of TF_chain ;
 - 5) Delete the sn_i^p in the Y set;
 - 6) endif
 - 7) endif
 - 8) Endfor
-

Algorithm 1. Redundant blocks are eliminated.

the Beta distribution is used as the basic trust value probabilistic statistical model for calculating the power terminal. The large probability density function of Beta distribution is.

$$\text{Beta}(\alpha, \beta) = \frac{\psi(\alpha, \beta)}{\psi(\alpha)\psi(\beta)} \tau^{\alpha-1} (1 - \tau)^{\beta-1} \quad (2)$$

Among them, τ represents the possibility of power terminal behavior, $0 < \tau < 1$, $\alpha > 0, \beta > 0$.

In the power Internet of things, the BTF trust factor times are used to calculate the basic trust value. When the PT_i authentication is successful, $\alpha = btf_{i1} + 1$. Otherwise $\beta = btf_{i2} + 1$. Therefore, the basic trust value of PT_i can be calculated using the beta distribution function:

$$bt_i = \text{Beta}(\alpha + 1, \beta + 1), \quad (3)$$

Obviously, both α and β are integers. The expected value of the beta distribution function is derived as

$$E[\text{Beta}(\alpha, \beta)] = \frac{\alpha}{\alpha + \beta} \quad (4)$$

Further calculation of bt_i :

$$bt_i = \frac{btf_{i1} + 1}{btf_{i1} + btf_{i2} + 2} \quad (5)$$

However, the static trust evaluation feature of bt_i has a certain hysteresis for the response of the power terminal's failure threat behavior. If the PT_i is lost, the bt_i cannot be attenuated quickly, which will bring the attacker multiple opportunities to implement threats. As a subjective state, trust can change with the dynamic changes of user interaction experience, time, and other factors. Using static trust for calculation will gradually deviate the recommendation results from

the real state. Therefore, it is necessary to introduce an abnormal decay factor to improve the basic trust value and enable dynamic trust evaluation. In our suggested approach, we combine the delegates selection problem from the permissioned blockchain and the learner's selection problem from federated learning into a single problem of combinational node selection.

In the power Internet of things, the ATF trust factor count is used to calculate the abnormal attenuation factor, which assists the abnormal monitoring and certification of the zero-trust engine. The outlier factor for PT_i is calculated as follows:

$$af_i = \frac{1}{s \times \max(A_i)} \sqrt{\sum_{q=1}^s (atf_{iq})^2} \quad (6)$$

Among them, $A_i = \{atf_{i1}, \dots, atf_{iq}, \dots, atf_{is}\}$ is the set of ATF trust factor times of PT_i ; $\max(A_i)$ is the maximum value in the set, which is used to normalize abnormal factors.

After introducing the abnormal attenuation factor af_i , the dynamic trust value of PT_i can be calculated as:

Input: Ω ;

Output: ar_i .

- 1) Calculate the basic trust value bt_i ;
 - 2) If $bt_i < \sigma$ then
 - 3) $ar_i = 0$ authentication failed;
 - 4) Else
 - 5) Calculate the dynamic trust value dt_i ;
 - 6) If $dt_i < \sigma$ then
 - 7) $ar_i = 0$ authentication failed;
 - 8) Else
 - 9) Calculate the terminal trust value t_i ;
 - 10) Endif
 - 11) If $t_i < \sigma$ then
 - 12) $ar_i = 0$ authentication failed;
 - 13) Else
 - 14) $ar_i = 1$, the authentication is successful;
 - 15) Endif
 - 16) Endif
-

Algorithm 2. Power terminal trusted certification.

$$dt_i = \begin{cases} \left\{ \frac{btf_{i1} + 1}{btf_{i1} + btf_{i2} + 2} - \frac{1}{s \times \max(A_i)} \sqrt{\sum_{q=1}^s (atf_{iq})^2}, dt_i > af_i \right. \\ 0, dt_i \leq af_i \end{cases} \quad (7)$$

At the same time, using the risk perception mechanism of the zero-trust client, real-time perception and collection of the STF trust factor times, calculation of the sensitive factor sf_i , and timely response to the mutation behavior of the lost terminal, that is,

$$sf_i = \begin{cases} \sigma, \forall (stf_{iu})_{u=1}^v \geq 1 \\ 0, \text{all}(stf_{iu})_{u=1}^v = 0 \end{cases} \quad (8)$$

When the times of any STF trust factor are greater than or equal to 1, the assignment of sf_i is the trust threshold σ . After the malicious code or instructions on the compromised terminal are removed, the control right is regained, and the security is strengthened, the number of times the STF trust factor of the PT_i is cleared. The zero-trust client enters a new real-time perception and collection stage. Federated learning for personalized recommendation systems has several benefits, one of which is the ability for the system to adapt to the preferences and routines of many users, producing suggestions that are more accurate and pertinent [23]. In order to handle important challenges like data privacy, data security, data access rights, and access to heterogeneous data, federated learning enables different players to develop a single, strong machine learning model without sharing data. Moving users' interests between pertinent contexts is the primary goal of cross-context suggestions. Cross-context recommendation systems may employ meta-learning or federated learning strategies. The main objective of cross-context suggestions is to move users' interests between relevant contexts. Federated learning or meta-learning approaches may be used by cross-context recommendation systems.

The sensitive factor sf_i is to quickly adjust the dynamic trust value of PT_i below the threshold value. Therefore, the final terminal trust value can be calculated as.

$$t_i = \begin{cases} |dt_i - sf_i|, & \forall (stf_{iu})_{u=1}^v \geq 1 \\ dt_i, & \text{all}(stf_{iu})_{u=1}^v = 0 \end{cases} \quad (9)$$

PE calls the trust factor set Ω stored on TF_chain and obtains the authentication result ar_i about PT_i through Algorithm 2.

4.3 Signcryption transfer of edge authentication information

After the authentication is successful, the PA generates the authorization credential ac_i according to the authentication result ar_i submitted by the PE and sends it to the PEP to form the edge authentication information $mi = (PT_i \parallel ar_i \parallel ac_i \parallel ts_i)$. Among them, ts_i is the certification letter timestamp of the message. If the edge authentication information passed to the PDC is in the clear text state, the attacker

can hijack and tamper with it during transmission. The encrypted communication of mi can ensure the security and integrity of edge authentication information.

As shown in Fig. 5, the parties involved in encryption transmission include the critical generation center (KGC), PEP and PDC. Among them, KGC is responsible for randomly generating keys; PEP encrypts and signs mi ; PDC is responsible for receiving and verifying the correctness of the ciphertext. The realization of the encryption transfer scheme includes five processes.

1. System parameter generation: KGC selects a security parameter K to generate two large prime numbers, p and q . G is an additive cyclic group of order q , and P is a generator of G . Define the hash function $H_1 : \{0, 1\}^* \times G \times G \rightarrow Z_q^*$, $H_2 : G \times G \times G \rightarrow \{0, 1\}^*$, $H_3 : \{0, 1\}^* \times \{0, 1\}^* \times G \times G \times G \times G \rightarrow Z_q^*$. KGC randomly selects $x \in Z_q^*$ as the master key. The public key of the computing system $P_{Pub} = x \times P$. KGC keeps the master key x and discloses the system parameters $params = (p, q, P, P_{Pub}, H_1, H_2, H_3)$.
2. Partial key generation: PEP randomly selects $x_{PEP} \in Z_q^*$ and calculates the public key $P_{PEP} = x_{PEP} \times P$. Then, the identity information ID_{PEP} and public key P_{PEP} of PEP are transmitted to KGC through a secure channel. KGC randomly selects $r_{PEP} \in Z_q^*$ and calculates part of the public key $R_{PEP} = r_{PEP} \times P$, part of the private key $s_{PEP} = r_{PEP} + h_1^{PEP} \times x$. And pass $\langle R_{PEP}, s_{PEP} \rangle$ to PEP through a secure channel. Among them, $h_1^{PEP} = H_1(ID_{PEP}, R_{PEP}, P_{PEP})$.
3. User key generation: PEP verifies whether part of the private key is valid by calculating $s_{PEP} \times P = R_{PEP} + h_1^{PEP} \times P_{Pub}$. If correct, the public and private keys of PEP are $PK_{PEP} = (R_{PEP}, P_{PEP})$, $SK_{PEP} = (s_{PEP}, x_{PEP})$.
4. Signcryption: Similarly, the public and private keys of PDC can be calculated as $PK_{PDC} = (R_{PDC}, P_{PDC})$, $SK_{PDC} = (s_{PDC}, x_{PDC})$. There are five steps in the encryption process of PEP for edge authentication information mi :

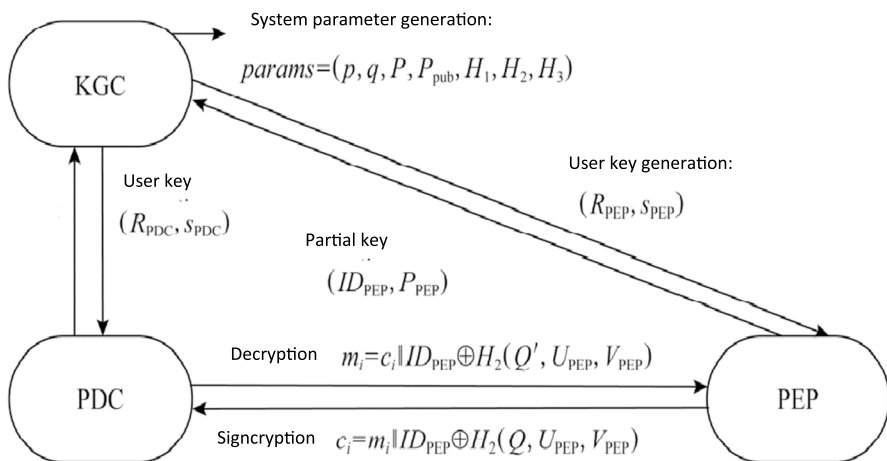


Fig. 5 Signcryption for edge authentication information

- (i) PEP randomly selects $u_{\text{PEP}} \in Z_q^*$, $v_{\text{PEP}} \in Z_q^*$, and calculates $U_{\text{PEP}} = u_{\text{PEP}} \times P$, $V_{\text{PEP}} = v_{\text{PEP}} \times P$.
 - (ii) Calculate $Q = u_{\text{PEP}} \times (h_1^{\text{PDC}} \times P_{\text{Pub}} + R_{\text{PDC}}) + v_{\text{PEP}} \times P_{\text{PDC}}$.
 - (iii) Calculate $c_i = m_i \parallel \text{ID}_{\text{PEP}} \oplus H_2(Q, U_{\text{PEP}}, V_{\text{PEP}})$.
 - (iv) Calculate $h_2^{\text{PEP}} = H_3(c_i, \text{ID}_{\text{PEP}}, U_{\text{PEP}}, V_{\text{PEP}}, R_{\text{PEP}}, P_{\text{PEP}})$.
 - (v) Calculate $\text{sig}_{\text{PEP}} = x_{\text{PEP}} + h_2^{\text{PEP}}(u_{\text{PEP}} + s_{\text{PEP}})$. PEP generates signcrypted ciphertext $\delta_i = (U_{\text{PEP}}, V_{\text{PEP}}, c_i, \text{sig}_{\text{PEP}})$, and sends it to PDC.
5. Decrypt the encryption: PDC receives the ciphertext δ_i , de-signcrypts it, and verifies the correctness of the ciphertext. There are 4 steps in the decryption process:

- (i) $h_2^{\text{PEP}} = H_3(c_i, \text{ID}_{\text{PEP}}, U_{\text{PEP}}, V_{\text{PEP}}, R_{\text{PEP}}, P_{\text{PEP}})$.
- (ii) Calculate $Q' = U_{\text{PEP}} \times s_{\text{PDC}} + x_{\text{PDC}} \times V_{\text{PEP}}$.
- (iii) Calculate $m_i = c_i \parallel \text{ID}_{\text{PEP}} \oplus H_2(Q', U_{\text{PEP}}, V_{\text{PEP}})$.
- (iv) Calculate $\text{sig}_{\text{PEP}} \times P = X_{\text{PEP}} + h_2^{\text{PEP}} \times (U_{\text{PEP}} + R_{\text{PEP}} + h_1^{\text{PEP}} \times P_{\text{Pub}})$.

The following proves the correctness of the encryption scheme.

Verify the authenticity of the message. Then, according to formula (10), we can know that $Q' = Q$, according to which the plaintext $m_i \parallel \text{ID}_{\text{PEP}} = c_i \oplus H_2(Q', U_{\text{PEP}}, V_{\text{PEP}})$ can be decrypted correctly.

$$\begin{aligned}
 Q' &= U_{\text{PEP}} \times s_{\text{PDC}} + x_{\text{PDC}} \times V_{\text{PEP}} = u_{\text{PEP}} \times s_{\text{PDC}} \times P \\
 &\quad + v_{\text{PEP}} \times x_{\text{PDC}} \times P = (R_{\text{PDC}} \times h_1^{\text{PDC}} \times x) \times u_{\text{PEP}} \\
 &\quad \times P + v_{\text{PEP}} \times P_{\text{PDC}} = u_{\text{PEP}} \times R_{\text{PDC}} + u_{\text{PEP}} \times h_1^{\text{PDC}} \\
 &\quad \times P_{\text{Pub}} + v_{\text{PEP}} \times P_{\text{PDC}} = u_{\text{PEP}} \times (R_{\text{PDC}} + h_1^{\text{PDC}} \times P_{\text{Pub}}) \\
 &\quad + v_{\text{PEP}} \times P_{\text{PDC}} = Q
 \end{aligned} \tag{10}$$

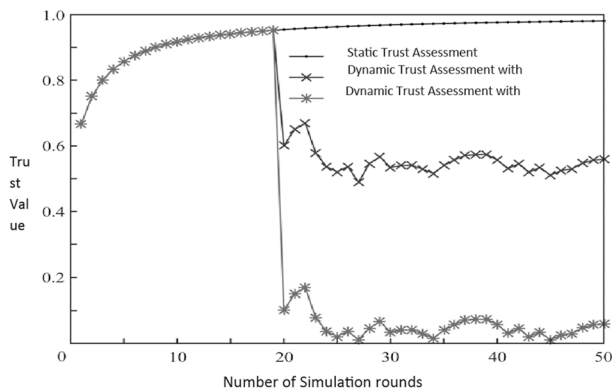
Verify the authenticity of the signed message, the verification formula is

$$\begin{aligned}
 \text{sig}_{\text{PEP}} \times P &= X_{\text{PEP}} + h_2^{\text{PEP}} \times (U_{\text{PEP}} + R_{\text{PEP}} + h_1^{\text{PEP}} \times P_{\text{Pub}}) \\
 &= x_{\text{PEP}} \times P + h_2^{\text{PEP}} \times P \times (u_{\text{PEP}} + r_{\text{PEP}} + h_1^{\text{PEP}} \times x) \\
 &= (x_{\text{PEP}} + h_2^{\text{PEP}} \times (u_{\text{PEP}} + r_{\text{PEP}} + h_1^{\text{PEP}} \times x)) \times P \\
 &= (x_{\text{PEP}} + h_2^{\text{PEP}} \times (u_{\text{PEP}} + s_{\text{PEP}})) \times P
 \end{aligned} \tag{11}$$

In this scheme, PEP calculates $\text{sig}_{\text{PEP}} = x_{\text{PEP}} + h_2^{\text{PEP}}(u_{\text{PEP}} + s_{\text{PEP}})$ for signature. KGC and PEP jointly generate PEP's private key SK_{PEP} . Suppose the attacker can obtain the identity information of the PEP. In that case, calculating the signcryptor's private key is a problem of the elliptic curve discrete logarithm, and it is difficult for the attacker to forge its signature. Therefore, this scheme can resist the tampering of edge authentication information and ensure safe transmission to PDC.

Table 3 Simulation parameters

Parameter	Meaning	Default value
N	Number of power terminals	2010
E	Number of edge areas	5, 15, 25
Z	Number of zero-trust engines	10–30
T_s	Number of simulation rounds	50
m	Lost terminal ratio	35%
σ	trust threshold	0.51

**Fig. 6** Variation of trust value for a compromised terminal

5 Simulation analysis

5.1 Simulation environment settings

In this paper, MATLAB is used to build an experimental platform. The proposed scheme is simulated and analyzed to verify the suppression effect against the compromised terminal threat and the implementation efficiency of the edge zero-trust security model. The simulation environment parameters are set as shown in Table 3.

5.2 Simulation analysis and results of combating the compromised terminal threat

In the zero-trust security architecture, the trust value is a sufficient basis for realizing the rapid authentication of power terminals. Through 50 rounds of experimental simulation, the changes in the trust value of the compromised terminal under the three trust evaluation schemes are observed. As shown in Fig. 6, in the first 20 rounds, the power terminal was not lost, and the trust values of the three schemes tended to increase; in the 20th round, the power terminal suddenly lost control, and

the observation results changed. Experiments are used to verify the efficacy of self-attention-based federated learning as well as its benefits in dealing with data sparsity and the cold start problem. The peripheral informal interface serves as the connection between the IoT terminal and the sensing device. It is made up of three main components: a central processing module, an external communication interface, and a peripheral perception (sensing) interface. Security flaws and hazards exist with smart terminals, and manufacturers have inserted backdoors into the firmware of related terminals. For recognizing both distinct local heads for each client and a typical data representation across users, we provide a completely new federated learning architecture and methodology. For each update of the representation, our method uses the shared computational resources across clients to carry out a number of local adjustments regarding the low-dimensional local parameters. A self-attention module and an embedding layer make up the Self-attention based federated learning (SAFL) model, a cross-context transfer network and a decision network. We offer an entirely new federated learning architecture and algorithm for identifying both unique local heads for each client and a typical data representation across users. Our method employs the shared computational resources among clients to perform several local adjustments about the low-dimensional local parameters for each update of the representation. Furthermore, we define in-depth experimental results that demonstrate how our approach outperforms rival personalized federated learning systems in a variety of settings. Due to its essential functions in protecting contemporary vital infrastructures, such as the identification of injected false data and the diagnosis of transmission line failures in smart grids, Internet of things (IoT) anomaly detection is significant. Researchers have put forth a number of detection strategies supported by machine learning (ML) methodologies. Due to its benefits of privacy preservation and decreased latency, federated learning (FL), a promising distributed ML paradigm, has recently been used to enhance detection performance. However, efficiency, robustness, and security issues still plague current FL-based techniques.

For the static trust assessment scheme, the hysteresis of the trust value response makes the compromised terminal always succeed in authentication but causes the continuous growth of the trust value. The dynamic trust evaluation scheme introducing the abnormal factor af_i can quickly respond to the failure of the power terminal. Still, the attenuation of the trust value is limited and fluctuates slowly around the threshold value. As the number of rounds increases, it will attenuate below the threshold but may give too many threats to the lost terminal. The introduction of abnormal factor af_i and sensitive factor sf_i dynamic trust evaluation scheme can deal with the sudden loss of the power terminal in time and quickly attenuate its trust value below the threshold value. When the trust value of the compromised terminal is lower than the threshold value, the authentication cannot be successful, thus losing the opportunity for malicious threats. An edge area is randomly selected as an example, in which the number of power terminals is 200 and the proportion of lost terminals. It is 30%. Large collections of labelled data and neural network topologies that automatically learn characteristics from the data are used to train deep learning models. Simply said, training a model entails learning (deciding) appropriate values for each weight and bias from labeled samples. Empirical risk minimization is the

technique by which a machine learning algorithm constructs a model in supervised learning by analyzing several examples and looking for a model that minimizes loss.

As shown in Fig. 7, the static trust evaluation scheme cannot attenuate the trust value of the compromised terminal, and the number of malicious threats in the edge region cannot be suppressed. The dynamic trust evaluation scheme that introduces the abnormal factor af_i is challenging to attenuate the trust value of the lost terminal below the threshold value. A zero-trust engine should be installed around the dense power terminals to counter the possible threat of compromised terminals given the authentication requirements in the scenario of widespread terminal access to the power Internet of things. The requirements for authentication in the case of numerous terminals accessing the power IoT. Additionally, deploying zero-trust in the delay-sensitive power IoT scenario is problematic if the deployment of the zero-trust engine is too far away from the power terminal. This is because it is simple to postpone the authentication time.

There is difficulty in suppressing the number of malicious threats. However, the dynamic trust evaluation scheme that introduces the abnormal factor af_i and the sensitive factor sf_i can block the lost terminal in time, and the number of malicious threats is quickly cleared in the event of an emergency.

Further, observe the detection of the lost terminal in the edge region. As shown in Fig. 8, the dynamic trust evaluation scheme that introduces the abnormal factor af_i and the sensitive factor sf_i offers a reasonable detection rate of the lost terminal. It is cut off and detected in time when the signs of the lost terminal appear.

5.3 Efficiency simulation results and analysis of edge zero-trust model

The edge deployment scheme can significantly reduce the zero-trust processing load of cloud PDC. As shown in Fig. 9, the zero-trust processing load under the centralized deployment scheme increases linearly with the increase in resource access requests. The power Internet of things has long been a major target of network assaults as a key infrastructure relating to the national economy and people's

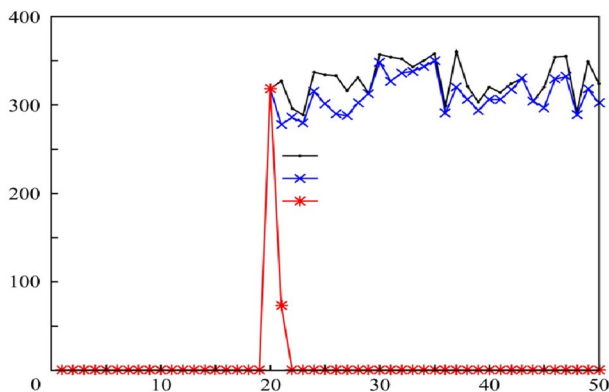


Fig. 7 Suppression of malicious threats

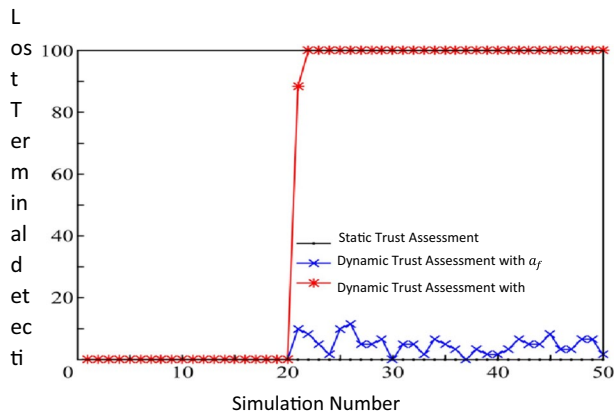


Fig. 8 Detection rate of compromised terminals

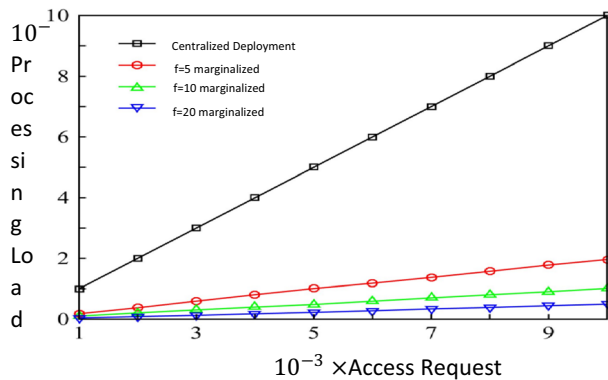


Fig. 9 Comparison of processing overload

livelihood. The types and quantity of terminals connected to the power Internet of things have greatly increased as a result of the widespread use of emerging technologies such as “Big Cloud, IoT, Smart Chain Edge” in the power industry. However, after distributing to edge regions, the more edge regions are divided, the less zero-trust processing load each edge region undertakes.

Two schemes can be adopted when deploying the zero-trust engine in each edge area: physical configuration or virtualized configuration. As shown in Fig. 10, virtualizing the PE, PA and PEP core components of the zero-trust engine to an edge server, compared with the solution of physically configuring an edge server for each core component, effectively reducing the need for edge servers. Federated learning is vulnerable to model poisoning attacks. The challenge comes from the fact that the centralized server cannot view the training data provided by the clients and therefore

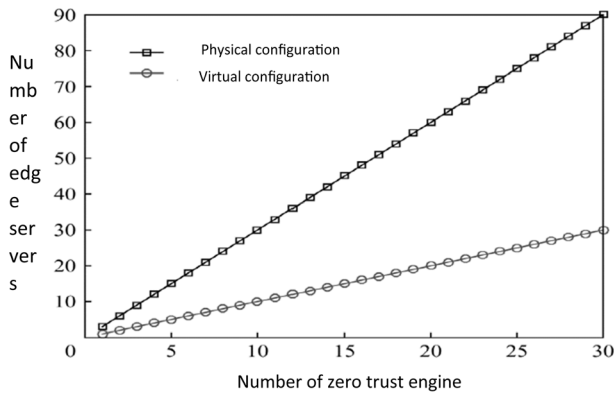


Fig. 10 Edge server consumption

it is impossible to validate the clients' changes. To address the problems, we present a unique model poisoning attack defense method for federated learning in this study. Our method checks each client's training results using a validation dataset to determine whether these clients' updates are suitable for aggregation. Attacks occur when malicious clients exchange gradients with the FL server during the training process, directly altering the learning rule. The power Internet of things is now much more widely known thanks to the power industry's ongoing investment in information technology. Attackers can steal confidential information from the power industry system or cause damage by using the compromised terminal as a launching pad to attack the network. An edge zero-trust model is used to overcome the bottleneck of centralized zero-trust deployment of huge power terminal access.

Finally, the generated edge authentication information is transmitted to the PDC using the encryption scheme, effectively preventing tampering. To prove the computational efficiency and lightness of the encryption scheme in this paper, a comparison is made with the latest typical encryption scheme.

The computational efficiency of the encryption scheme mainly depends on the point multiplication E_m and point addition E_a operations on the elliptic curve. Combined with the specific measurement results ($E_m = 0.4847$ ms, $E_a = 0.0021$ ms), it can be seen from Table 4 that the scheme in this paper has apparent advantages in computational efficiency. Therefore, the encryption scheme in this paper can ensure

Table 4 Comparison of computation efficiency

Program	Signcryption time	Decryption time	Total time
Reference scheme	$4E_m + 2E_a \approx 1.9431$	$7E_m + 4E_a \approx 3.4015$	5.3445
Reference scheme	$5E_m + 2E_a \approx 2.4275$	$7E_m + 2E_a \approx 3.3975$	5.8245
Reference scheme	$6E_m + 2E_a \approx 2.9125$	$8E_m + 6E_a \approx 3.8905$	6.8025
Signcryption scheme of this paper	$5E_m + 2E_a \approx 2.4275$	$5E_m + 4E_a \approx 2.4315$	4.8595

the safe transmission of edge authentication information and, at the same time, has better computational efficiency and lightness.

6 Conclusion

An edge zero-trust model is proposed to aim at the centralized deployment pressure of zero-trust facing massive power terminal access. The distributed multi-point deployment of the zero-trust engine is carried out around the power terminal, and the real-time collection of trust factors and the blockchain storage scheme is designed. Introducing anomalous factors and through dynamic trust evaluation, it can deal with the impulsive behavior of the lost terminal and rapidly attenuate its trust value. Around the power terminal, a distributed multi-point deployment of the zero-trust engine is made, and a real-time trust factor collecting and blockchain storage scheme are established. It can handle the impulsive behavior of the lost terminal and quickly reduce its trust value by introducing anomalous factors and using dynamic trust evaluation. To guarantee the security of prompt blocking of the edge authentication information threatened by the lost terminal to the cloud, a lightweight sign encryption approach is used. The simulation results show the edge of zero-confidence proposed in this paper. The arbitrary model effectively suppresses the threat of the lost terminal and has a better detection rate of the lost terminal. The future research work is to deeply analyze the characteristics of east–west traffic and design an effective and rapid detection method for encrypted malicious traffic sent by the compromised terminal inside the power Internet of things and method to further enhance the suppression effect of the lost terminal threat. In recommendation systems, federated learning has been utilized to protect user privacy. Because recommendation systems can train suggestion models without using actual user data by gathering intermediate parameters as a substitute, user privacy is considerably improved when they are utilized in federated learning contexts. The effectiveness of self-attention based federated learning and its advantages in overcoming data sparsity and the cold start problem are confirmed by studies.

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Author contributions AMAS and AR wrote the main manuscript text and provide conceptualization, methodology, software; MSC and LLCC prepared figures and provide data curation; MS provides validation, investigation and edited the manuscript. All authors reviewed the manuscript.

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Declarations

Conflict of interest The author declares no conflict of interest.

Ethical approval Not applicable.

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